Question 1:

Find $\int_0^2 \frac{5}{\sqrt{x}} dx = \lim_{b \to 0^+} \int_b^2 \frac{5}{\sqrt{x}} dx$

a) The indefinite integral $\int \frac{5}{\sqrt{x}} dx = 5\int x^{-1/2} dx = ?$

b) $\int_0^2 \frac{5}{\sqrt{x}} dx = \lim_{b \to 0^+} \int_b^2 \frac{5}{\sqrt{x}} dx =$ ____________________________

Question 2:

Find $\int_0^3 \frac{7}{(x-1)^2}dx = \lim_{b\to 1^-}\int_0^b \frac{7}{(x-1)^2}dx + \lim_{c\to 1^+}\int_c^3 \frac{7}{(x-1)^2}dx$

a) $\lim_{b\to 1^-}\int_0^b \frac{7}{(x-1)^2}dx =$? __________

b) $\lim_{c\to 1^+}\int_c^3 \frac{7}{(x-1)^2}dx \quad =$? __________

c) $\int_0^3 \frac{7}{(x-1)^2}dx =$? __________

Question 3:

Find $\int_5^\infty \frac{1}{x^4}dx = \lim_{b\to\infty}\int_5^b \frac{1}{x^4}dx$

a) The indefinite integral $\int \frac{1}{x^4}dx = ?$

b) $\int_5^\infty \frac{1}{x^4}dx = \lim_{b\to\infty}\int_5^b \frac{1}{x^4}dx =$ ____________________

Question 4:

Find $\int_1^\infty \frac{1}{\sqrt[4]{x}}dx = \lim_{b\to\infty}\int_1^b \frac{1}{\sqrt[4]{x}}dx$

a) The indefinite integral $\int \frac{1}{\sqrt[4]{x}}dx = ?$

b) $\int_1^\infty \frac{1}{\sqrt[4]{x}}dx = \lim_{b\to\infty}\int_1^b \frac{1}{\sqrt[4]{x}}dx =$ ______________________

Question 5:

Find $\int_0^{\infty} xe^{-2x}dx = \lim_{b\to\infty}\int_0^{b} xe^{-2x}dx$

a) The indefinite integral $\int xe^{-2x}dx = ?$

b) $\int_0^{\infty} xe^{-2x}dx = \lim_{b\to\infty}\int_0^{b} xe^{-2x}dx =$ ______________________

Question 6:

Find $\int_0^\infty x^2 e^{-x}\,dx = \lim_{b\to\infty}\int_0^b x^2 e^{-x}\,dx$

a) The indefinite integral $\int x^2 e^{-x}\,dx = ?$

b) $\int_0^\infty x^2 e^{-x}\,dx = \lim_{b\to\infty}\int_0^b x^2 e^{-x}\,dx =$ ______________________

Question 7:

Find $\int_{-\infty}^{\infty} \frac{3}{25+x^2}dx = \lim_{b\to-\infty}\int_{b}^{0} \frac{3}{25+x^2}dx + \lim_{c\to\infty}\int_{0}^{c} \frac{3}{25+x^2}dx$

a) $\lim_{b\to-\infty}\int_{b}^{0} \frac{3}{25+x^2}dx =$ _?__________

b) $\lim_{c\to\infty}\int_{0}^{c} \frac{3}{25+x^2}dx =$ _?__________

c) $\int_{-\infty}^{\infty} \frac{3}{25+x^2}dx =$ _?__________